

INDUSTRIAL TRAINING PROGRAM ON ROBOTICS AND AI

Mini Project Report

Industrial 2-Finger Gripper

CAD Design & Static Stress Simulation

Submitted by

Bipasha Deka

Tool Used: SolidWorks Premium 2023

Date: April 2026

1. Problem Statement

The objective of this project is to design and analyze a 2-finger parallel jaw industrial gripper suitable for robotic applications. The gripper must be fully modeled using parametric CAD tools, include fingers, base, and actuation interface, be capable of holding a payload between 5 kg and 10 kg, and pass static stress analysis under applied gripping forces.

The final design has been validated using static structural simulation in SolidWorks Simulation.

2. Design Concept

2.1 Mechanism Description

The gripper uses a rack-and-pinion mechanism to achieve parallel jaw motion. A central pinion gear meshes simultaneously with two racks mounted on linear sliders. When the actuator rotates the pinion, both racks move in opposite directions along the X-axis, carrying the fingers symmetrically toward or away from the center. This ensures perfectly parallel gripping motion at all times.

2.2 Working Principle

- Actuator (motor) rotates the central pinion gear
- Pinion teeth mesh with both Rack L and Rack R simultaneously
- Rack L moves left while Rack R moves right (or vice versa)
- Each rack is bolted to a slider block riding on linear rails
- Each finger is bolted to the front face of its slider
- Both fingers move symmetrically toward the object and grip it

2.3 Color Scheme

- Fingers: Yellow (Aluminium 6061-T6)
- Base plate and sliders: Blue (Aluminium 6061-T6)
- Racks and pinion: As per SolidWorks Toolbox default

3. Material Properties

Material selected: Aluminium Alloy 6061-T6

Justification: Lightweight, high strength-to-weight ratio, excellent machinability, and corrosion resistance — ideal for industrial robotic grippers.

Property	Value	Unit
Elastic Modulus (E)	68,900	MPa
Poisson's Ratio	0.33	-
Yield Strength	275	MPa

Ultimate Tensile Strength	310	MPa
Density	2,700	kg/m ³
Thermal Conductivity	167	W/(m·K)

4. CAD Part Dimensions

4.1 Component Summary

Component	Dimensions (mm)	Source
Pinion Gear	Module=1, Z=20, PCD=20, W=20	SolidWorks Toolbox
Rack (×2)	Module=1, Z=30, L=110, W=20	SolidWorks Toolbox
Finger (×2)	80 × 50 × 30	Modeled
Slider (×2)	54 × 55 × 30	Modeled
Base Plate	200 × 120 × 10	Modeled
Linear Rails (×2)	200 × 15 × 8	Feature on Base Plate

4.2 Key Design Parameters

Parameter	Value
Gear Module	1 mm
Pinion Pitch Circle Diameter	20 mm
Maximum Jaw Opening	100 mm
Rack Travel per Revolution	62.83 mm
Actuator Revolutions for Full Open	1.6 turns
Minimum Safety Factor	1.5

5. Gripping Force Calculation

5.1 Design Load

Maximum payload: 10 kg (worst case)

Weight of object: $W = m \times g = 10 \times 9.81 = 98.1 \text{ N}$

5.2 Required Gripping Force

Assumptions:

- Coefficient of friction between rubber grip pad and object: $\mu = 0.4$

- Safety factor: $SF = 1.5$
- Two fingers sharing the load equally

Formula: $F_{grip} = (W \times SF) / (2 \times \mu)$

$F_{grip} = (98.1 \times 1.5) / (2 \times 0.4)$

$F_{grip} = 147.15 / 0.8$

$F_{grip} = 184 \text{ N per finger (368 N total)}$

5.3 Required Actuator Torque

Torque = $F_{grip} \times (\text{Pinion PCD} / 2)$

Torque = $184 \times (0.020 / 2)$

Required Torque = 1.84 N·m

6. Assembly Description

6.1 Assembly Sequence

- Base plate fixed to ground as reference component
- Two linear rails extruded on top face of base plate
- Left and right sliders mated to rails using coincident mates — free to slide along X axis
- Rack L mated to top face of Slider L — moves together as one unit
- Rack R mated to top face of Slider R — moves together as one unit
- Pinion positioned at center using plane coincident mates — free to rotate
- Rack-Pinion mechanical mate applied between pinion and both racks
- Finger L bolted to front face of Slider L
- Finger R bolted to front face of Slider R

6.2 Degrees of Freedom

Component	Motion Allowed	Motion Constrained
Base Plate	None (fixed)	All 6 DOF
Slider L	Translate along X	Y, Z translation + all rotation
Slider R	Translate along X	Y, Z translation + all rotation
Pinion	Rotate about Z axis	All translation
Rack L	Translate along X (with Slider L)	All other DOF
Rack R	Translate along X (with Slider R)	All other DOF
Finger L	With Slider L	Fixed to Slider L
Finger R	With Slider R	Fixed to Slider R

6.3 CAD Assembly Screenshots

The following screenshots show the complete gripper assembly from multiple views as modeled in SolidWorks 2023.

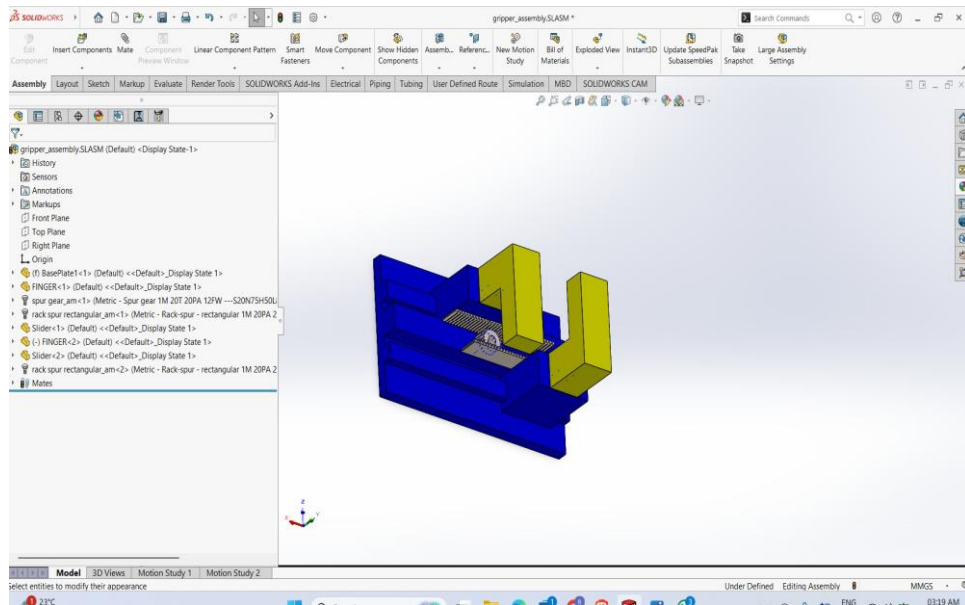


Figure 1 — Isometric (Trimetric) view of complete gripper assembly

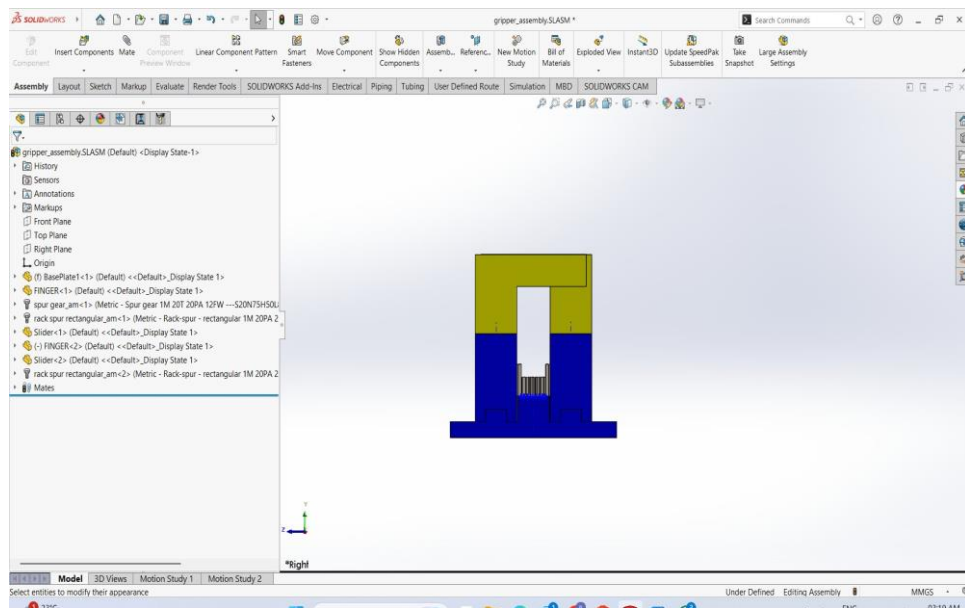


Figure 2 — Right side view showing finger and slider arrangement

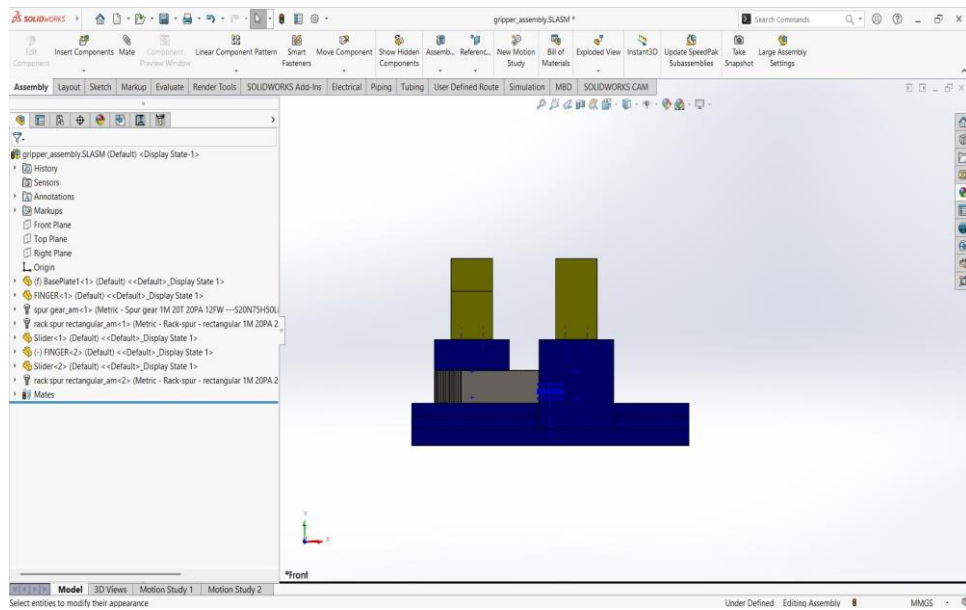


Figure 3 — Front view showing parallel jaw configuration

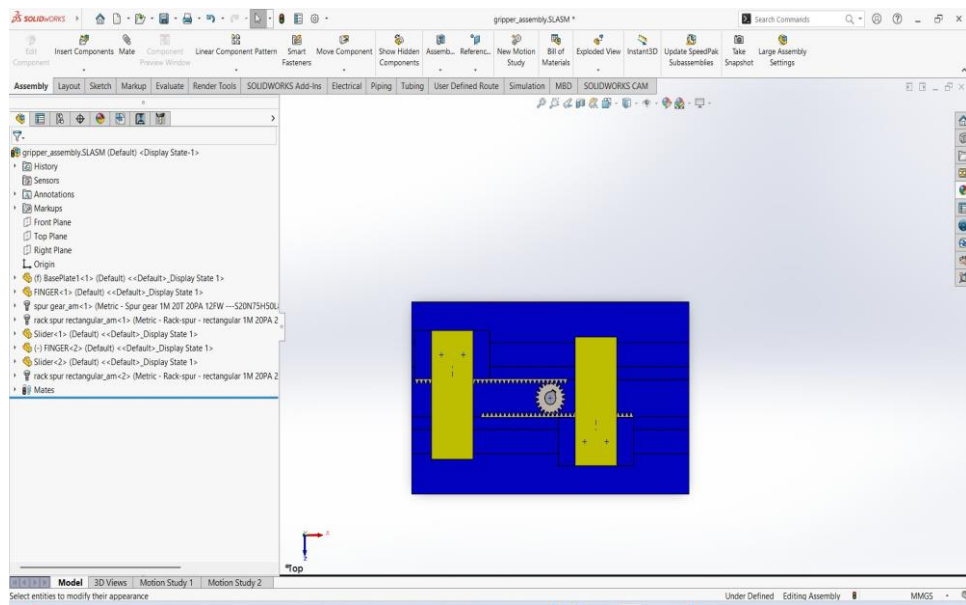


Figure 4 — Top view showing rack-and-pinion mechanism with both fingers

7. Static Stress Simulation

7.1 Simulation Setup

- Software: SolidWorks Simulation (Static Study)
- Material: Aluminium 6061-T6 applied to all parts
- Fixture: Fixed geometry on bottom face of base plate
- Load: 184 N applied normal to inner grip face of each finger
- Mesh: Fine mesh, element size approximately 2-3 mm

7.2 Simulation Results

Parameter	Result	Limit	Status
Max Von Mises Stress	3.838 MPa	< 184 MPa	PASS
Max Displacement	0.00768 mm	< 0.5 mm	PASS
Equivalent Strain	3.288×10^{-5}	< 0.001	PASS
Calculated Safety Factor	71.8	> 1.5	PASS
Yield Strength (Al 6061-T6)	275 MPa	-	-

7.3 Results Interpretation

Von Mises Stress: The maximum stress of 3.838 MPa is significantly below the allowable stress of 184 MPa (yield strength divided by safety factor of 1.5). The design is extremely safe with respect to yielding.

Displacement: Maximum deformation of 0.00768 mm is negligible — well within the 0.5 mm acceptable limit. The gripper maintains its shape under full gripping load with virtually no deformation.

Safety Factor: The achieved safety factor of 71.8 far exceeds the minimum requirement of 1.5. This indicates that the Aluminium 6061-T6 material is more than adequate for the applied loads. The design could theoretically handle loads up to 47 times the design load before yielding.

Equivalent Strain: The strain value of 3.288×10^{-5} confirms that the material remains well within its elastic region under the applied gripping forces. No plastic deformation occurs.

8. Conclusion

A 2-finger parallel jaw industrial gripper has been successfully designed and analyzed using SolidWorks. The gripper uses a rack-and-pinion mechanism driven by a central actuator to achieve symmetric parallel jaw motion. The design meets all project requirements:

- Fully parametric CAD model with all components modeled and assembled correctly
- Rack-and-pinion mechanism provides true parallel jaw motion
- Gripping force of 184 N per finger safely holds 5-10 kg payload with safety factor of 1.5
- Static stress simulation confirms maximum stress of 3.838 MPa — well below the 184 MPa limit
- Maximum displacement of 0.00768 mm confirms rigid structure
- Overall safety factor of 71.8 exceeds the minimum requirement of 1.5

The design is structurally sound, mechanically correct, and suitable for industrial robotic applications. The use of Aluminium 6061-T6 provides an excellent balance of strength, weight, and machinability for this gripper application.

References

- SolidWorks 2023 Documentation and Help Files
- Shigley's Mechanical Engineering Design, 10th Edition
- Aluminium 6061-T6 Material Data Sheet — ASM International
- Robo AI Industrial Training Program on Robotics and AI — Assignment Guidelines

Industrial_Gripper_Design_BipashaDeka

Submitted to: Robo AI — Industrial Training Program on Robotics and AI